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SUMMARY

Full-stack developer who takes production platforms from architecture through deployment and operations. Sole designer of a multi-venue hospitality operations platform on Azure, now building an end-to-end encrypted federated messaging platform. The independent work runs on one vision: infrastructure that leaves communities in control of their own tools.

TECHNICAL SKILLS

Languages & Frameworks: PHP, Laravel, Livewire, JavaScript, TypeScript, React, Node.js, Rust, Python, MySQL, HTML/CSS
Security & Protocols: MLS (RFC 9420), OpenMLS, libsignal, PQXDH, Double Ratchet, end-to-end encryption, threat modeling, adversarial and mutation testing
Platform & Infrastructure: Azure deployment and optimization, axum, SQLite, database design, systems architecture, multi-tenant platforms, REST API integration, CI/CD, Git
Integrations: Toast POS, Easol, Embed, Viator, Signal protocol
AI-Assisted Delivery: Claude Code and Codex for spec-driven implementation, test-guided generation, and iterative review, with architecture and design decisions owned directly

SELECTED PROJECTS

MYCO FEDERATION

JUL 2026 - PRESENT

CREATOR AND SOLE DEVELOPER: ACTIVE DEVELOPMENT

RUST, LIBSIGNAL, OPENMLS (RFC 9420), AXUM, SQLITE

Federated, end-to-end encrypted communications platform giving community organizers a secure Discord-shaped structure. Phases 0 through 5 of a 10-phase roadmap complete in seven weeks: identity, direct messages, channels, and the server core all run and are demonstrated in full.

- Built the protocol core in Rust on libsignal and OpenMLS: PQXDH and Double Ratchet for direct messages, MLS with client-signed membership for channels, and a purpose-built OpenMLS provider in place of the upstream test provider
- Made membership integrity a compile-time guarantee rather than a runtime check, so a forged commit is structurally unable to reach the step that merges it and a hostile server cannot add itself to a channel
- Shipped the server core (registration, key directory, MLS delivery service, mailbox queues, WebSocket fan-out, epoch recovery) on storage-as-truth with per-entity linearizers, so the server sequences commits while holding no group keys
- Hardened the unauthenticated surface before any deployment: per-source rate limiting, per-sender mailbox quotas, fail-closed TLS startup that refuses to bind rather than fall back to plaintext
- Proved every security property by executable test, not code review: each defense is deliberately broken to confirm exactly the right guardian test fails. 94% line coverage on the kernel, 1,000-member channels measured end to end, and libsignal and OpenMLS each confined to one adapter module by a CI isolation check
- Audited the completed phases against Signal, Matrix, Wire, Mastodon, and Discord's production MLS deployment, plus the governing RFCs, before building the server on top of them

ALEXANDRIA

2021 - PRESENT

CREATOR AND SOLE DEVELOPER: WORKING PROTOTYPE

REACT, LARAVEL, EAV MODEL, LLM INTEGRATION

Worldbuilding and creative structuring platform built around a problem no existing tool solves: how writers capture unstructured ideas as they arrive, then organize them as the world takes shape.

- Architected the platform on an entity-attribute-value model so users define their own entity schemas instead of conforming to a fixed template, letting one system serve very different creative projects with no customization
- Built an LLM integration layer that reads each user's custom schema at runtime and generates content conforming to that structure rather than producing generic output
- Migrated the frontend to React on a production-scale codebase to work directly with modern patterns

SIGNAL ACCESS MANAGER

JAN 2026 – MAR 2026

CREATOR AND SOLE DEVELOPER: PILOTED + LIMITED DEPLOYMENT

SIGNAL PROTOCOL

Secure communications and room automation for community rapid-response networks, built on Signal's open-source infrastructure. Ten phases and 2,000+ tests, concept to live pilot in eight weeks, with certificate-based verification, ZIP code adjacency alerting for geo-aware coordination, and silent emergency modes for high-risk scenarios.

THE EMPOWERED PUBLIC

JUL 2026 - PRESENT

CREATOR AND SOLE DEVELOPER: EARLY DEVELOPMENT

Toolkit that helps residents prepare for town hall meetings on data center development, replacing grandstanding with direct questions that demand direct answers on resource stewardship and promised jobs.

PROFESSIONAL EXPERIENCE

SWINGERS: CRAZY GOLF CLUB

MAR 2022 – APR 2026

LEAD PLATFORM DEVELOPER

APR 2025 – APR 2026

*Sole developer of the **Swingers Hub**, a full-stack internal platform replacing fragmented Excel-based operations across multiple hospitality venues.*

- Architected and deployed a Laravel/Livewire platform on Azure with 106 database tables, 211 routes, and 61 reusable service classes, integrating Toast, Easol, Embed, and Viator for real-time operational data
- Shipped 53 production releases and 1,100+ commits solo, delivering in eight months a Phase 1 scope estimated at 12 to 18 months for a four-to-six-person team
- Cut discrepancy research from 30–60 minutes to under 5 minutes, and nightly cash reconciliation by 20–30 minutes per venue, by replacing manual Excel workflows with automated API-driven pipelines
- Separated queue and notification logic into independent deployments to hold reliability on a cost-efficient resource tier
- Designed multi-venue architecture that onboarded a new location with zero additional custom development, validating franchising readiness

APPLICATION SUPPORT ANALYST

MAR 2022 – APR 2025

- Automated operational data extraction, reducing hourly processing time from 12 minutes to 30 seconds
- Led first systemic improvements to Toast POS configuration, standardizing workflows and reducing errors across venues
- Made the business case for a central internal platform during the Easol onboarding, leading to the Hub initiative

EARLIER CAREER

Sodexo USA, Assistant Manager of Guest Services (2021): Founded the 8 West Concierge Program for VIP clientele and launched the Lobby Concierge Program, consolidating service functions under one department during a hiring freeze.

JetBlue Airways, Inflight Crewmember (2014–2021): Departmental safety champion for two years, representing inflight on enterprise-wide safety committees during major operational changes.

Walt Disney World Resort, Trek Coordinator / Payroll Process Analyst / Guest Relations Trainer (2007–2014): Established the Coordinator Help Desk at Animal Kingdom, the central call center for guest impacts, saving over 25,000 labor hours annually.

EDUCATION

Northwood University (3+1 Program)	2005 – 2008	B.S. Business Administration
Delta College (3+1 Program)	2005 – 2008	Associate in Business Studies

Code samples and private repositories available upon request. Selected projects, including Myco, Alexandria and Signal Access Manager, are maintained in private repositories due to the personal and/or security-sensitive nature of the work.